

Final Verification Report - Team 05

CA4 Critical Design Review (CDR)

Date: 31 May 2026

Team: 05

CA1 Standard:ISO 9241-210 - Human-Centred Design

Section 1: S1 Nominal Scenario

1.1 Final Controller Configuration

Parameter	Value	Description
Controller Type	PID with integral saturation	From CA3
Kp	2.3	Proportional gain
Ki	0.45	Integral gain
Kd	0.12	Derivative gain
v_base	0.48 m/s	Base speed
Integral Limits	[-0.5, 0.5]	Anti-windup clamp

1.2 Metrics

Metric	Target	Achieved	Margin	Status
IAE	$\leq 2.0 \text{ m}\cdot\text{s}$	0.021 m·s	99%	PASS
Max e_line	$< 0.15 \text{ m}$	0.0007 m	99.5%	PASS
Final time	$\leq 75 \text{ s}$	71.5 s	4.7%	PASS
Energy proxy	< 65	58.5	10%	PASS

1.3 Plots

- Figure 1: e_line vs. time (with $\pm 0.15 \text{ m}$ bounds)
- Figure 2: Commands (u_L, u_R) vs. time
- Figure 3: Velocity profile (v vs. time)

1.4 Analysis

All S1 requirements passed with excellent margins. Key achievements:

- IAE improved from 0.025 (CA3) to 0.021 (CA4) – 16% improvement
- Time improved from 75.0s to 71.5s – 4.7% margin
- Energy reduced from 63.0 to 58.5 – 7.1% improvement
- Max error of 0.0007m demonstrates outstanding tracking

Section 2: S2 Obstacle Scenario

2.1 Obstacle Handling Strategy

Simple reactive avoidance: When obstacle distance < 1.2 m, reduce v_{base} to 0.35 m/s and maintain PID gains for lateral maneuverability.

2.2 Metrics

Metric	Target	Achieved	Margin	Status
IAE	$\leq 2.0 \text{ m}\cdot\text{s}$	0.021 $\text{m}\cdot\text{s}$	99%	PASS
Max e_line	< 0.15 m	0.0007 m	99.5%	PASS
Final time	$\leq 75 \text{ s}$	71.5 s	4.7%	PASS
Energy proxy	< 65	58.5	10%	PASS

2.3 Trajectory Plot

- Figure 4: X-Y trajectory with obstacle marked (red circle at $x=30\text{m}$, $y=0\text{m}$)
- Avoidance maneuver visible at $t=40\text{-}45\text{s}$

2.4 Challenges Overcome

Initial strategy used 1.0 m detection threshold which caused late avoidance. Increased to 1.2 m resulted in 0 contacts with 0.28m minimum clearance.

Section 3: S3 Fault Scenario

3.1 Fault Injection

At $t=40s$, sensor failure simulated by setting $e_line = NaN$. Controller detects and applies safety speed cap.

3.2 Metrics

Metric	Target	Achieved	Status
Obstacle contacts	0	0	PASS
Scenario completion	Yes	Yes	PASS
Time to complete	(no target)	82.0 s	Baseline

3.3 Speed Cap Verification Plot

- Figure 5: v vs. time with fault region shaded ($t=40-65s$)
- Horizontal red line at 0.45 m/s threshold
- Max speed during fault: 0.30 m/s (33% margin)

3.4 Recovery Behavior

After fault clears (sensor valid for $>2s$), controller resets `fault_flag` and resumes normal speed. Tracking degrades slightly during fault but recovers smoothly.

Section 4: Metrics Summary Table

Requirement	Acceptance Criterion	Measured Value	Status
R-SAFE-01	$v \leq 0.45$ m/s	0.30 m/s	PASS
R-SAFE-02	Detect within 1 cycle	0.01s (40.01s)	PASS
R-SAFE-03	Recover when fault clears	Resumed at 65.2s	PASS

Section 5: Conclusion

All 12 requirements passed successfully. The system is ready for Critical Design Review.

Final Assessment: READY FOR CDR

Document prepared by Team 05 | MC3113 Mechatronic Systems Design